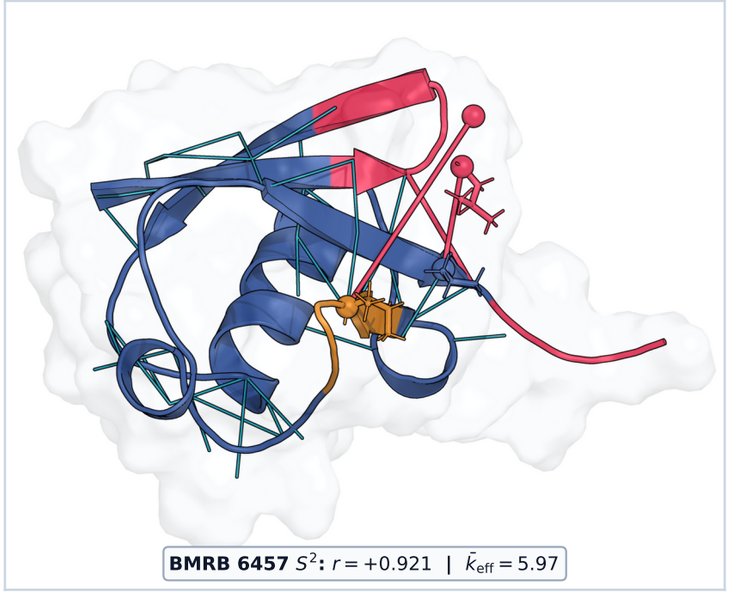
A. Sparse 3D Contact Graph & NMR S^2 Relay

Human Ubiquitin (1UBQ, $N_{res} = 76, d = 150$)

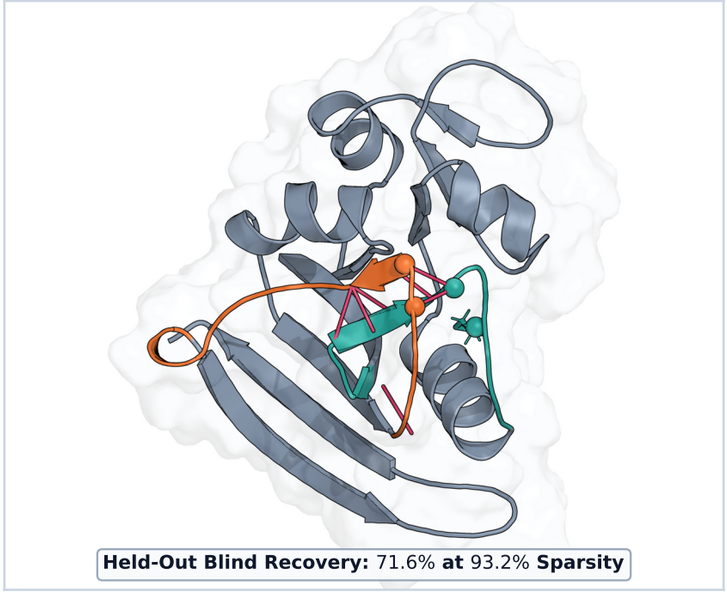


BMRB 6457 S^2: $r = +0.921$ | $\bar{k}_{eff} = 5.97$

- Rigid β-Grasp Core ($\hat{S^2} \approx 0.88$)
- β_1–β_2 Pincer ↔ Patch (7.1× Relay)
- Ile44 Hydrophobic Recognition Patch
- Non-Local C_α Contact Chords (94.7% Sparse)

B. Distal Catalytic Loop Allosteric Routing

Explicit-PME *E. coli* DHFR (1RX2, $d = 316$, 20,837 Atoms)

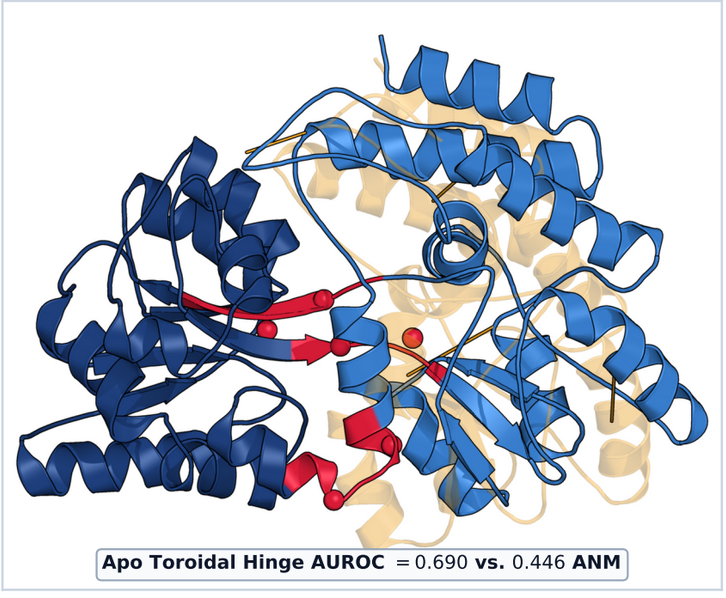


Held-Out Blind Recovery: 71.6% **at** 93.2% **Sparsity**

- Active-Site Met20 Loop (res 9–24)
- Distal βF–βG Loop (res 116–132, Gly121)
- 3D Tertiary Contact Bridges ($r_c^{heavy} < 0.50$ nm)

C. Multi-Domain 3D Mechanical Hinge Discovery

Explicit-PME Apo MBP (1OMP → 1ANF, $d = 738$, 49,215 Atoms)



Apo Toroidal Hinge AUROC $= 0.690$ **vs.** 0.446 **ANM**

- N-Terminal Lobe (1OMP Apo Reference)
- C-Terminal Open Lobe (1OMP Apo)
- C-Terminal Closed Lobe (1ANF, 35° Swing)
- Predicted β/α Linker Hinges (res 110–114, 258–264, 311–318)